UNIVERSAL BOUNDEDNESS OF A QUASI-FINITE MORPHISM OVER A DENSE OPEN EXTENDS TO A NEIGHBOURHOOD OF THE GENERIC POINTS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Degree stratification. lemma-quasi-compact-reasonable-stratification (carded L26710). The degree stratification of U over X.
- Scheme-theoretic density. Morphisms of Spaces, lemma semi-diagonal (cross-chapter, named). Density of the closure piece.

Source reference. Original reference 03BM: focused-seeds/geometry/S9-decent-generically-finite/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ACK: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4630–4630. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03M5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3910–3910. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ABY: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3963–3963. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01WN: focused-seeds/geometry/S9-decent-generically-finite/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02NW: focused-seeds/geometry/S9-decent-generically-finite/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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Source reference. Original reference 03GN: focused-seeds/geometry/S9-decent-generically-finite/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AAY: focused-seeds/geometry/S9-decent-generically-finite/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03ZQ: focused-seeds/geometry/S9-decent-generically-finite/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ABN: focused-seeds/geometry/S9-decent-generically-finite/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BAQ: focused-seeds/geometry/S9-decent-generically-finite/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume that f is quasi-separated of finite type. Let $y \in |Y|$ be a point of codimension 0 on Y. The following are equivalent:*

1. *the space $|X_k|$ is finite where $\mathrm{Spec}(k) \to Y$ represents y,*
2. *$X \to Y$ is quasi-finite at all points of $|X|$ over y,*
3. *there exists an open subspace $Y' \subset Y$ with $y \in |Y'|$ such that $Y' \times_Y X \to Y'$ is finite.*

If Y is decent these are also equivalent to

4. *the set $f^{-1}(\{y\})$ is finite.*

Proof. The equivalence of (1) and (2) follows from Lemma 0ACK (and the fact that a quasi-separated morphism is decent by Lemma 03M5).

Assume the equivalent conditions of (1) and (2). Choose an affine scheme V and an étale morphism $V \to Y$ mapping a point $v \in V$ to y. Then v is a generic point of an irreducible component of V by Properties of Spaces, Lemma 0BAQ. Choose an affine scheme U and a surjective étale morphism $U \to V \times_Y X$. Then $U \to V$ is of finite type. The morphism $U \to V$ is quasi-finite at every point lying over v by (2). It follows that the fibre of $U \to V$ over v is finite (Morphisms, Lemma 0AAY).

By Morphisms, Lemma 02NW after shrinking V we may assume that $U \to V$ is finite. Let

$$R = U \times_{V \times_Y X} U$$

Since f is quasi-separated, we see that $V \times_Y X$ is quasi-separated and hence R is a quasi-compact scheme. Moreover the morphisms $R \to V$ is quasi-finite as the composition of an étale morphism $R \to U$ and a finite morphism $U \to V$. Hence we may apply Morphisms, Lemma 02NW once more and after shrinking V we may assume that $R \to V$ is finite as well. This of course implies that the two projections $R \to V$ are finite étale. It follows that $V/R = V \times_Y X$ is an affine scheme, see Groupoids, Proposition 03BM. By Morphisms, Lemma 03GN we conclude that $V \times_Y X \to V$ is proper and by Morphisms, Lemma 01WN we conclude that $V \times_Y X \to V$ is finite. Finally, we let $Y' \subset Y$ be the open subspace of Y corresponding to the image of $|V| \to |Y|$. By Morphisms of Spaces, Lemma 03ZQ we conclude that $Y' \times_Y X \to Y'$ is finite as the base change to V is finite and as $V \to Y'$ is a surjective étale morphism.

If Y is decent and f is quasi-separated, then we see that X is decent too; use Lemmas 03M5 and 0ABY. Hence Lemma 0ACK applies to show that (4) implies (1) and (2). On the other hand, we see that (2) implies (4) by Morphisms of Spaces, Lemma 0ABN. □